

INCIDENT REPORT

INC-2026-003

Linux Cryptominer Payload Delivery via SFTP — Masquerading as sshd

STATUS: ● CLOSED

1. INCIDENT OVERVIEW

Ticket ID	INC-2026-003
Title	Linux Cryptominer Payload Delivery via SFTP — Masquerading as sshd
Analyst	Jacek Kaplon
Severity	CRITICAL
Detection Source	Wazuh Manager — Rule 100203 (SFTP Upload) + Rule 87105 (VirusTotal Alert)
Upload Time	2026-06-17 02:37:21 UTC — SFTP file upload detected
VT Alert Time	2026-06-17 02:37:23 UTC — VirusTotal confirmed malicious (2s latency)
Sensor Node	rpilab (agent.id: 017) 192.168.1.11
Classification	TRUE POSITIVE — Active Malware Delivery (Linux Cryptominer / Trojan)

2. THREAT ACTOR PROFILE

Source IP	49.173.65.19
Geolocation	Seodaemun-gu, Seoul, South Korea (lat: 37.5761 / lon: 126.9344)
Protocol	SSH / SFTP
Session ID	15f5710aa6b3
Sensor ID	9507f202fe70
UUID	32df9ab2-2fe7-11f1-a9b7-da8176cc5bca
data.eventid	cowrie.session.file_upload
Log source	/home/pi/cowrie/logs/cowrie.json

3. PAYLOAD ANALYSIS

Filename	sshd ■ masquerading as SSH daemon binary
File type	ELF 64-bit (x86-64, dynamically linked, UNIX System V)
File size	23.44 MB (24576000 bytes)
SHA-256	d666a49ef825163c40186b93a0858d14c31c7a2ee77c048c2106034557768b22
SHA-1	356f643865b5c23b6e591fa35afc6a32c1917bc3
MD5	edfc321265094254927e65ec16268c71
VT Detection	37 / 65 engines — MALICIOUS
Threat Label	trojan, miner Family: r002c0de925
First Submission	2024-01-15 16:36:37 UTC (known payload, not zero-day)

Last Submission	2025-02-12 20:09:54 UTC
Last Analysis	2025-08-21 02:07:18 UTC
VT Permalink	https://www.virustotal.com/gui/file/d666a49ef825163c40186b93a0858d14c31c7a2ee77c048c2106034557768b22

Selected AV verdicts:

- Kaspersky: HEUR:Trojan.Linux.Miner.gen
- ClamAV: Unix.Trojan.Miner-9993889-0
- ESET-NOD32: Linux/CoinMiner.ABF
- Microsoft: Trojan:Linux/Multiverse
- Elastic YARA: Linux_Generic_Threat
- AhnLab-V3: CoinMiner/Linux.Agent.30304472

4. ATTACK TIMELINE

02:37:21 UTC | Rule 100203 (sev 10) | cowrie.session.file_upload
| SFTP upload detected from 49.173.65.19 (Seoul, South Korea)
| filename: 'sshd' (masquerading as SSH daemon)
| SHA-256: d666a49ef825163c40186b93a0858d14c31c7a2ee77c048c2106034557768b22
| groups: post_exploitation, virus_detection
| MITRE: T1059 (Command Interpreter) + T1105 (Ingress Tool Transfer)

02:37:23 UTC | Rule 87105 (sev 12) | VirusTotal API enrichment complete
| data.virustotal.found: 1
| data.virustotal.positives: 37 / 65 engines
| data.virustotal.malicious: 1
| MITRE: T1203 (Exploitation for Client Execution)

Upload-to-VT-verdict latency: 2 seconds

5. INVESTIGATION & ROOT CAUSE

Attack Pattern: The threat actor gained SSH access to the honeypot and immediately used SFTP to upload a 23.44 MB ELF64 Linux cryptominer binary. The file was deliberately named sshd to masquerade as the legitimate OpenSSH daemon — a common persistence and defence evasion technique on Linux systems.

MITRE ATT&CK Chain:

- **T1105** — Ingress Tool Transfer: ELF binary delivered via SFTP
- **T1036** — Masquerading: binary named 'sshd' to blend with system processes
- **T1059** — Command and Scripting Interpreter: post-upload execution attempt
- **T1496** — Resource Hijacking: cryptominer family confirmed by 37 AV engines
- **T1203** — Exploitation for Client Execution: VirusTotal rule correlation

Threat Assessment: This is a known payload (first seen 2024-01-15, family r002c0de925) consistent with automated Linux cryptomining campaigns that scan for SSH endpoints with weak credentials, deploy a miner binary disguised as a system process, and attempt persistence via process name spoofing. The 23.44 MB binary size and ELF64 architecture confirm it is a standalone statically-linked miner compiled for x86-64 Linux targets.

6. CONTAINMENT & REMEDIATION

Containment	Cowrie emulation layer intercepted the upload — binary stored in isolated volume, no execution on host
Action Taken	Payload preserved for forensic analysis; SHA-256 confirmed against VirusTotal database
Recommended Action	Block 49.173.65.19 at edge firewall; submit hash to internal threat intel feed; implement Wazuh Active Response to auto-block on Rule 100203 trigger
Production Risk	NONE — honeypot architecture prevented any host-level execution

7. INDICATORS OF COMPROMISE (IoC) SUMMARY

Type	Value	Context
IP Address	49.173.65.19	SFTP upload source — Seoul, South Korea
Filename	sshd	Malicious ELF masquerading as SSH daemon
SHA-256	d666a49ef...7768b22	Linux cryptominer — 37/65 VT engines malicious
MD5	edfc3212...16268c71	VT confirmed hash
SHA-1	356f6438...1917bc3	VT confirmed hash
File size	23.44 MB (ELF64 x86-64)	Standalone cryptominer binary
Family	r002c0de925	Trojan/Miner family label (VirusTotal)
Session ID	15f5710aa6b3	Cowrie session — SFTP upload
Rule IDs	100203 + 87105	SFTP upload + VirusTotal alert
Geo	Seoul, South Korea (37.57/126.93)	GeoIP auto-enriched by Wazuh

8. ANALYST COMMENTS & LESSONS LEARNED

This incident represents the highest-severity capture in the current lab series — a real-world malware delivery event where an attacker successfully uploaded a known Linux cryptominer to the honeypot via SFTP. The naming convention (sshd) demonstrates deliberate defence evasion tradecraft, consistent with organised cryptomining campaigns targeting exposed Linux SSH endpoints globally.

Key observations:

- Upload-to-VT-verdict latency was **2 seconds** — automated pipeline operating at production speed
- Binary named sshd — attacker intended to disguise miner as legitimate system process for persistence and evasion
- Payload is a **known commodity tool** (first seen Jan 2024) — not targeted attack, part of automated mass-scanning campaign
- Cowrie preserved the full 23.44 MB binary for forensic analysis — hash immediately cross-referenced against 65 AV engines
- This event combined with INC-2026-002 confirms the honeypot is attracting multiple independent threat actors simultaneously

Next Step: Wazuh Active Response integration to automatically firewall source IPs upon Rule 100203 (SFTP upload) trigger — preventing repeated delivery attempts from confirmed malicious actors at the network edge.

9. REFERENCES & RELATED INCIDENTS

- INC-2026-001 — Automated Malware Triage: EICAR Pipeline Validation
- INC-2026-002 — Unauthorized SSH Access & Post-Compromise Reconnaissance (177.136.32.81, Brazil)
- MITRE ATT&CK: T1105, T1036, T1059, T1496, T1203
- VirusTotal report: [d666a49ef825163c40186b93a0858d14c31c7a2ee77c048c2106034557768b22](https://www.virustotal.com/gui/file/d666a49ef825163c40186b93a0858d14c31c7a2ee77c048c2106034557768b22)
- Full project: github.com/jacekkaplon/The-Littlefish-Hunter